

Saumil Singhal

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EDUCATION

Vellore Institute of Technology, Bhopal

B.Tech in Computer Science Engineering (Artificial Intelligence & Machine Learning)

Bhopal, India

2023 – 2027

EXPERIENCE

Ethara AI

Remote

LLM Post-Training Intern

Mar 2026 – Apr 2026

- Debugged and diagnosed hallucinations, logical inconsistencies, and edge-case failures in **LLM** outputs, performing systematic **root cause analysis** against structured evaluation frameworks.
- Engineered corrective interventions via **prompt engineering** and iterative experimentation, applying a hypothesis-test-analyze loop to resolve incorrect or inefficient model behavior.
- Collaborated cross-functionally with engineers and stakeholders under tight timelines, communicating technical findings clearly to deliver reliable, production-facing outputs.

Bharat Heavy Electricals Limited (BHEL)

Haridwar, India

Full-Stack Developer Intern

May 2026 – June 2026

- Architected a **relational database schema (SQLite/PostgreSQL)** and built **REST API**-backed **Node.js/Express** services powering a **React** frontend for the Smart Material & Inventory Management Platform (SMIMP) – live at bhel-project-gray.vercel.app.
- Engineered a **predictive forecasting engine** into the backend to model material demand from historical data, exposing results to the frontend via **REST API** endpoints.
- Owned the system end-to-end – schema design, API architecture, debugging, and UI – independently translating stakeholder requirements into shipped, reliable features.

PROJECTS

Real-Time Hand Gesture Detection System | *Python, OpenCV, MediaPipe, Computer Vision* Jul 2026

[\[GitHub\]](#)

- Developed a real-time computer vision application using **OpenCV** and **MediaPipe** to detect and track **21 hand landmarks** from live webcam video streams with low-latency processing.
- Designed an image analysis pipeline involving **frame acquisition**, **RGB preprocessing**, ML-based landmark detection, **feature extraction**, and real-time visualization of detected hand keypoints.
- Built a modular Python implementation for continuous hand tracking, demonstrating practical applications in **gesture recognition**, **human-computer interaction**, virtual reality, and assistive technologies.

Credit Card Approval Model | *Python, TensorFlow, React, Flask, Scikit-learn*

May 2025 – Jun 2025

[\[GitHub\]](#) [\[Live Demo\]](#)

- Trained and evaluated multiple **ML models** on financial applicant data to drive automated **risk-decisioning** predictions, analyzing the key factors behind approval outcomes.
- Architected and deployed a full-stack decision-support system end-to-end, integrating a **Flask REST API** backend with a **React** dashboard for real-time predictions.
- Shipped a live, publicly accessible demo on GitHub, delivering the project from raw data to a working deployed product with an open, reviewable codebase.

Stock Market Trend Predictor | *Python, TensorFlow, Pandas, NumPy, Keras*

Apr 2025 – Jun 2025

[\[GitHub\]](#) [\[Live Demo\]](#)

- Built an end-to-end forecasting system using **LSTM** models on live **yFinance API** data to predict market trends.
- Analyzed and debugged multiple forecasting architectures, evaluating performance against historical market data to improve prediction accuracy.
- Deployed an interactive **Streamlit** dashboard, delivering the model as a usable, live analytics tool.

SKILLS

Programming Languages: Python, Java, SQL

Machine Learning & AI: Machine Learning, Deep Learning, TensorFlow, Keras, Scikit-learn, Predictive Modeling, Neural Networks, Model Training & Evaluation

Computer Vision & Image Processing: Computer Vision Fundamentals, Digital Image Processing, Image Classification, Feature Extraction, Image Segmentation (Fundamentals), Image Enhancement, Object Detection (Fundamentals)

Data Science: Pandas, NumPy, Data Preprocessing, Data Analysis, Data Visualization

Frameworks & Technologies: Flask, FastAPI, React.js, Node.js, Express.js, REST APIs, PostgreSQL, SQLite, MongoDB

Developer Tools: Git, GitHub, VS Code, Streamlit

CERTIFICATIONS

Java Foundation Associate – Oracle 